

OpenShift Scenarios (Infrastructure Administration)

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Scenario: Worker Node Unresponsive in OpenShift Cluster

First, check the node status with `oc get nodes`. SSH into the node to investigate CPU, memory, or disk utilization (`top`, `df -h`). If it's a network issue, ensure connectivity and restart kubelet (`systemctl restart kubelet`). For persistent failures, cordon and drain the node (`oc adm drain <node>`) and reintroduce it (`oc adm uncordon <node>`). Verify pod rescheduling and monitor logs (`journalctl -u kubelet`).

Scenario: Disk Pressure on Master Node

Check node status with `oc describe node <node>` for disk pressure. Run `df -h` to find large files. Clean up unused Docker images (`podman image prune -a`), old logs, and temporary files. Reduce log retention if needed (`/etc/kubernetes/logrotate.conf`). Consider increasing disk size for persistent issues.

Scenario: Cluster Certificate Expiry

Check certificate expiration using `oc get configmaps -n openshift-config-managed`. If nearing expiration, renew with the `oc adm certificate renewal` commands (`oc adm certificate approve`). For kubelet certificates, restart kubelet services. Backup the current configuration and regenerate certificates as required.

Scenario: Pods Stuck in ContainerCreating State

Inspect the pod's event logs using `oc describe pod <pod-name>`. Check for issues with Persistent Volumes or image pull secrets. If related to storage, verify the PV status and reclaim policies. For image pull issues, update the correct secret in the namespace (`oc set image-lookup`).

Scenario: API Server Down in OpenShift Cluster

Verify the status of API server pods (`oc get pods -n openshift-kube-apiserver`). Check logs for errors (`oc logs <pod>`). Restart the API server pods. Ensure etcd is healthy as it supports the API server (`oc get pods -n openshift-etcd`). If etcd is unhealthy, restore it from backup.

Scenario: Persistent Volume (PV) Not Bound to Persistent Volume Claim (PVC)

Check PVC status with `oc get pvc`. If it remains in a Pending state, inspect events with `oc describe pvc <pvc-name>`. Ensure the storage class exists and supports dynamic provisioning. For static provisioning, verify that PV and PVC have matching storage size, access modes, and labels. Adjust and recreate if needed.

Scenario: Node Not Ready State

Run `oc get nodes` to identify nodes in NotReady state. Check kubelet logs (`journalctl -u kubelet`). Investigate issues such as disk pressure, network errors, or insufficient CPU/memory. Restart kubelet or network services. If unresolved, reboot the node and monitor for recovery.

Scenario: Image Pull Back-off Error

Inspect the pod logs (`oc logs <pod-name>`). Ensure the image exists in the registry and the correct image pull secret is available. Check Docker or Podman image status on the node. If the image is private, create or update the pull secret using `oc create secret docker-registry`.

Scenario: Cluster Operators Degraded

Run `oc get co` to identify degraded operators. Investigate with `oc describe co <operator>`. Check logs and underlying pods (`oc logs <pod>`). Common causes include insufficient resources or node failures. Restart affected pods or redeploy the operator.

Scenario: High CPU Usage on API Server

Monitor API server usage (`oc adm top pod -n openshift-kube-apiserver`). Check logs for excessive API calls (`oc logs`). Identify offending users or applications. Consider applying rate limits. Scale API server replicas (`oc scale deployment`) for load balancing.

Scenario: Failed Node Eviction during Maintenance

Use `oc adm drain <node> --force --ignore-daemonsets` to evict all pods. Ensure there are sufficient resources on other nodes for pod rescheduling. Investigate stuck pods and delete them if necessary (`oc delete pod <pod>`). Verify pod migration using `oc get pods -o wide`.

Scenario: etcd Disk Space Full

Check etcd pod logs and disk usage on etcd nodes (`df -h`). Reduce etcd snapshot retention and defrag etcd (`oc exec -n openshift-etcd <etcd-pod> -- etcdctl defrag`). Backup and clean old snapshots. For persistent issues, expand disk space.

Scenario: Prometheus Storage Full

Monitor Prometheus storage usage (`oc get pvc -n openshift-monitoring`). If full, consider increasing the PVC size or deleting older data. Configure retention policies (`prometheus.yaml`) to reduce data retention.

Scenario: Route Not Accessible

Verify the route status with `oc get routes`. Check if the router pod is running (`oc get pods -n openshift-ingress`). Ensure the service backend is accessible. For external access, confirm DNS resolution and load balancer settings.

Scenario: Pod Cannot Communicate with External Services

Verify network policies with `oc get networkpolicy`. Ensure the required ports and protocols are open. Check the pod's DNS resolution (`nslookup <service>`). If using an external proxy, validate the proxy settings (`oc set env`).

Scenario: CronJob Fails to Execute

Check the CronJob schedule (`oc get cronjob <name>`) and last execution status. Inspect related pods (`oc logs <pod-name>`). Ensure there are sufficient node resources. Validate the CronJob's syntax and time zone configuration.

Scenario: Ingress Controller Pod CrashLoopBackOff

Inspect logs of the ingress controller (`oc logs <pod-name> -n openshift-ingress`). Check for certificate issues or conflicts in route configuration. Restart the pod. If necessary, redeploy the ingress controller operator.

Scenario: Authentication Failure with OAuth Server

Verify the OAuth pod status (`oc get pods -n openshift-authentication`). Check logs for error messages (`oc logs`). Ensure correct identity provider configuration in the cluster (`oc get oauth`). Restart the OAuth pod if necessary.

Scenario: Application Pods Not Scaling

Check the Horizontal Pod Autoscaler (`oc get hpa`). Ensure resource limits and requests are configured properly in the pod spec. If CPU or memory usage is below the target, adjust the threshold in the HPA definition.

Scenario: Failed Cluster Upgrade

Review the upgrade status (`oc get clusterversion`). Check logs for failed components. Identify blocking cluster operators with `oc get co`. Resolve issues and retry the upgrade. Backup etcd before retrying a major upgrade.

Scenario: Service Discovery Not Working

Verify service configuration with `oc describe svc <service>`. Check the DNS resolution (`nslookup <service-name>`). Ensure kube-dns pod is running (`oc get pods -n openshift-dns`). Inspect network policies and firewalls for blocking traffic. Restart kube-dns pod if necessary.

Scenario: Pod-to-Pod Communication Fails Across Namespaces

Check network policies in both namespaces (`oc get networkpolicy`). Ensure no restrictive policies block traffic. Verify that both pods are in Running state. Use `oc exec` to test communication. Allow traffic by modifying or adding network policies.

Scenario: NodePort Service Not Accessible Externally

Ensure the NodePort service is correctly defined (`oc describe svc <service>`). Check firewall rules on worker nodes to allow NodePort range (30000–32767). Verify service IP is reachable. Restart kube-proxy if necessary.

Scenario: Load Balancer Route Not Working

Inspect the route status (`oc get route`). Check logs for ingress controller pods. Ensure the load balancer is properly configured and connected to worker nodes. Validate DNS records for external access.

Scenario: Multus Secondary Network Not Configured

Verify Multus CNI configuration (`oc get net-attach-def`). Ensure the network attachment definition is correctly specified. Check the pod annotations for secondary network attachment. Redeploy the pod if required.

Scenario: High Network Latency in OpenShift Pods

Monitor network metrics (`oc adm top pods`). Identify pods with high latency. Use `tcpdump` and `ping` for network diagnosis. Check underlying network infrastructure and enforce QoS policies.

Scenario: IP Address Exhaustion in Cluster

Check the IP allocation status with `oc get clusternetwork`. Expand the CIDR range in the network configuration. Ensure no pods are holding unused IPs. If necessary, restart affected pods.

Scenario: VXLAN Connectivity Issues

Verify VXLAN tunnel status on nodes. Use `ovs-vsctl show` to check Open vSwitch status. Ensure MTU size matches across the cluster. Restart Open vSwitch and kubelet services if needed.

Scenario: External DNS Not Resolving in Pods

Check CoreDNS configuration (`oc get cm coredns -n openshift-dns`). Ensure external nameservers are correctly defined. Use `dig` or `nslookup` from the pod. Restart CoreDNS pods if necessary.

Scenario: Load Balancer Health Check Failing

Inspect the health check configuration on the load balancer. Check pod readiness probes and service endpoints (`oc describe svc`). Ensure the pods are healthy and running. Adjust health check intervals and thresholds.

Scenario: Prometheus Not Scraping Targets

Check the target status in Prometheus UI. Verify that the service monitor exists (`oc get servicemonitor`). Inspect Prometheus pod logs. Ensure the service is reachable and labels match.

Scenario: Alertmanager Failing to Send Alerts

Check the Alertmanager configuration (`oc get secret -n openshift-monitoring`). Verify email or webhook endpoints. Inspect logs for errors. Restart the Alertmanager pod if necessary.

Scenario: Grafana Dashboards Not Loading

Check Grafana pod status and logs (`oc logs grafana -n openshift-monitoring`). Ensure the data source is configured correctly. Verify that Prometheus is running and reachable.

Scenario: No Metrics for Custom Application

Ensure the application exposes metrics at the correct endpoint (`/metrics`). Create a service monitor with the appropriate labels. Check Prometheus for target discovery.

Scenario: Cluster Monitoring Operator Degraded

Run `oc get co monitoring`. Check logs for the monitoring operator pod. Inspect resource limits and increase them if needed. Restart the affected pods.

Scenario: Thanos Querier Not Responding

Verify Thanos Querier pod status (`oc get pods -n openshift-monitoring`). Check logs for errors. Ensure Prometheus and object storage are accessible. Restart Thanos Querier if required.

Scenario: Persistent Volume Full for Prometheus

Check PVC status (`oc get pvc -n openshift-monitoring`). Increase PVC size or delete old data. Configure retention policies to avoid recurrence.

Scenario: Missing Alerts for Node CPU Usage

Verify alert rules in Prometheus (`oc edit prometheusrule`). Ensure the node exporter is running and accessible. Adjust thresholds if necessary.

Scenario: Application Logs Not Forwarded to EFK Stack

Ensure Fluentd pods are running (`oc get pods -n openshift-logging`). Check the Fluentd configuration for log forwarding. Verify Elasticsearch and Kibana connectivity.

Scenario: Kibana Dashboard Not Showing Logs

Check the status of Elasticsearch and Fluentd. Verify index patterns in Kibana. Restart Elasticsearch pods if necessary and ensure sufficient disk space.

Scenario: PVC in Pending State

Inspect PVC events (`oc describe pvc`). Verify the storage class and provisioner. Ensure the storage backend is functional. Adjust the PVC size if required.

Scenario: ReadWriteOnce PVC Mounted by Multiple Pods

Ensure the PVC access mode is set to ReadWriteMany for shared access. Use a storage backend that supports RWX. Reconfigure the pods if necessary.

Scenario: Slow Read/Write Operations on PVC

Check the underlying storage performance. Monitor I/O usage (`iostat`). Adjust the storage class parameters for higher performance.

Scenario: NFS Mount Failure

Verify NFS server status and export configuration. Check pod logs for mount errors. Ensure network connectivity and proper permissions.

Scenario: OpenShift Data Foundation Pods Not Running

Inspect ODF operator logs. Check Ceph cluster health. Ensure sufficient resources for ODF components. Restart affected pods if necessary.

Scenario: Failed to Provision Persistent Volume

Check the storage class and provisioner logs. Verify the backend storage availability. Ensure the provisioner has sufficient permissions.

Scenario: Volume Snapshot Creation Fails

Verify the volume snapshot class exists (`oc get volumesnapshotclass`). Ensure the snapshot operator is running. Check snapshot logs for errors.

Scenario: Block Storage Not Recognized

Check the CSI driver status. Ensure the block device is available on the node. Verify node permissions to access the device.

Scenario: Disk Space Alerts Not Triggering

Ensure the monitoring stack is configured to monitor PVC usage. Create an alert rule for disk space thresholds. Test the alert manually.

Scenario: Data Loss After Pod Restart

Ensure data is stored on a Persistent Volume. Check if the PVC is still bound to the pod. Verify the volume mount path in the pod spec.

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